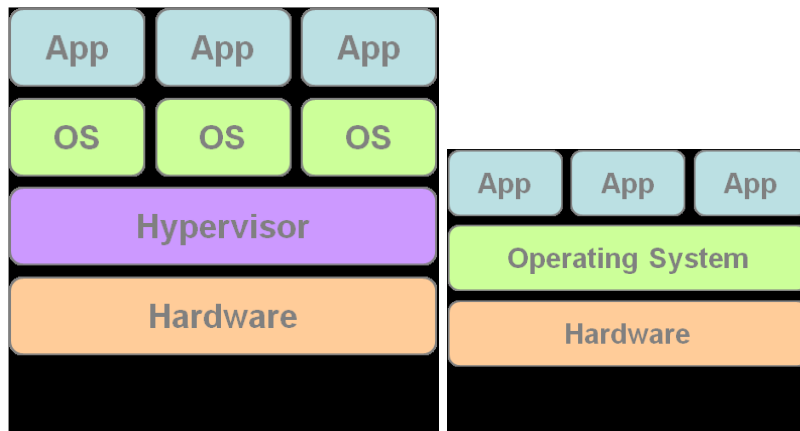


1. What are the system operation modes and how do they contribute to system security? [1 Marks]

User Mode and Kernel Mode

Code running in the kernel mode has unrestricted access to the hardware, while user mode restricts access to the CPU and memory

2. What distinguishes a Non-Virtual Machine from a Virtual Machine? Explain the contrast using a diagram. [2 Marks]



VM

Non-VM

3. Write three functions of class verifier within the JVM? [1 Marks]

Class verifier checks

.class file is valid Java byte code

- Does not overflow or underflow the stack
- Byte code does not perform illegal memory access

4. What is the relationship between operating systems and computer hardware? [1 Marks]

OS is responsible for managing all the computer's hardware and allows the computer hardware to communicate and operate with the application programs

5. What is the principal objective of time-sharing multiprogramming and batch multiprogramming? [1 Marks]

Batch Multiprogramming : Maximize processor use

Time Sharing: Minimize Response time

6. How does the OS kernel distinguish between user-mode and kernel-mode when processing system calls? [1 Marks]

Transition from user to kernel mode is represented by 0 or 1 bit

User Mode =1

Kernel Mode=0

[1 Marks]

7. Write only one disadvantage of passing parameters by registers versus passing parameters by stack. [1 Marks]

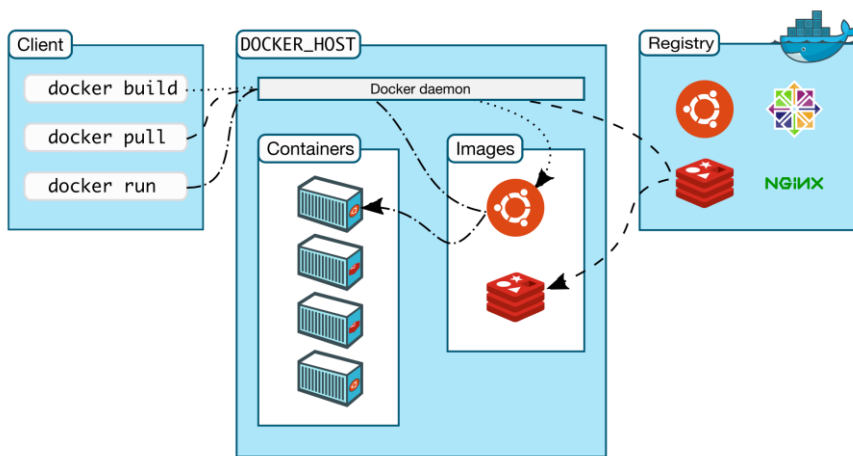
Registers have limited space may be in some cases may be more parameters than registers but in stack it does not limit the number or length of parameters being passed.

8. Compare batch processing and serial processing in the context of operating systems. Discuss the advantages and limitations of each approach [1.5 Marks]

In Batch Processing similar Types of jobs (Job control language) are Firstly Prepared and they are Stored on the Card. and that card will be Submit to the System for the Processing. The System then Perform all the Operations on the Instructions one by one. The **Main Problem is that the Jobs those are prepared for Execution must be the Same Type** and if a job requires for any type of Input then this will not be Possible for the user. And Many Time will be wasted for Preparing the Batch. The Batch Contains the Jobs and all those jobs will be executed without the user Intervention.

The Serial Processing Operating Systems are those which Perform all the instructions into a Sequence Manner or the Instructions those are given by the user will be executed by using the FIFO Manner means First in First Out. All the Instructions those are Entered First in the System will be Executed First and the Instructions those are Entered Later Will be Executed Later. Mainly the **Punch Cards** are used for this. In this all the Jobs **are firstly Prepared and Stored on the Card** and after that card will be entered in the System and after that all the Instructions will be executed one by One. But the **Main Problem is that a user doesn't interact with the System** while he is working on the System, means the user can't be able to enter the data for Execution.

9. Describe the key components of Docker's architecture and their interactions. How does each component contribute to Docker's functionality? [2.5 Marks]



10. Explain the role of system calls within the Unix operating system. How do they facilitate interaction between user space and kernel space? [1.5Marks]

Chapter 1 : Slide No 54

11. What are the advantages of using a layered system design in operating systems? [1.5 Marks]

Allows to built modular systems
Can exchange each layer
Facilitates testing
Look at each layer separately

12. What is JCL ? [1 Marks]

Job Control Language. It provides instructions to the monitor What compiler for different languages and to use and what to use

13. What is VMM ? [1 Marks]

Virtual machine Monitor a software layer providing the virtualization.

14. Write 2 types of system calls? [1 Marks]

Process Control
Device Management
Information maintenance
Communication
File Management

15. Explain the concept of JIT compilation in the context of the JVM. [1.5 Marks]
Just in Time (JIT) compiler helps to improve the performance of Java programs by compiling the byte codes into the machine code at runtime.

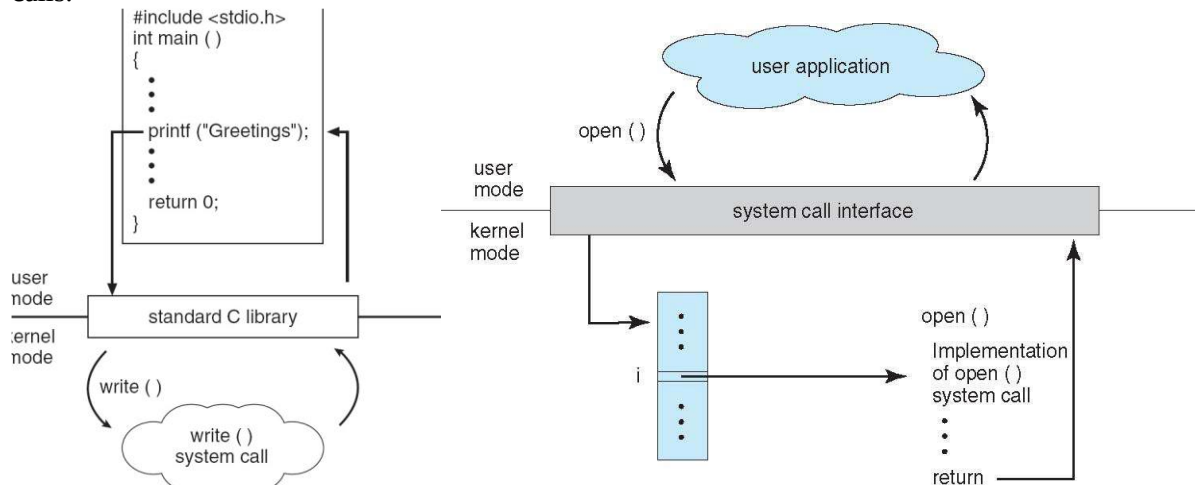
16. What is the relationship between operating systems and computer hardware? [1 Marks]
OS is responsible for managing all the computer's hardware and allows the computer hardware to communicate and operate with the application programs.

17. What is the primary difference between multiprogramming and uniprogramming ? [1 Marks]
In uniprogramming (at a time only one process can be in the main memory) one job prevents others from making progress. Processor must wait for I/O instruction to complete before proceeding. In multiprogramming (more than one process can be in a main memory at a time), When one job needs to wait for I/O, the processor can switch to the other job

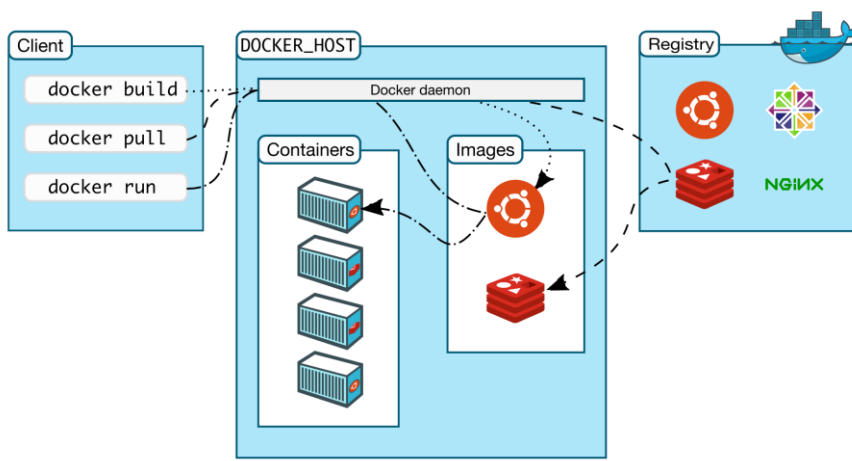
18. How does the operating system kernel distinguish between user-mode and kernel-mode when processing system calls? [1 Marks]
Transition from user to kernel mode is represented by 0 or 1 bit
User Mode =1
Kernel Mode=0

19. Write only one disadvantage of passing parameters by registers versus passing parameters by stack. [1 Marks]
Registers have limited space may be in some cases may be more parameters than registers but in stack it does not limit the number or length of parameters being passed.

20. Explain the process of how a user-level application interacts with the operating system kernel through system calls. [1.5 Marks]



21. Describe the key components of Docker's architecture and their interactions. How does each component contribute to Docker's functionality ? [2.5 Marks]



22. Discuss the evolution of operating system architectures from monolithic designs to more modular and layered approaches. What factors have driven this evolution? [2.5 Marks]
 Chapter 1: Introduction : Slide 52-56

Question no 1: Consider the following code

```
int main( )
{
    for (int i=0;i<2;i++)
    {
        If (! fork()){
            printf("CS");
            exit();
        }
        else
        {
            for (int j=0;j<3;j++)
            {
                if(fork () !=0){
                    printf("DS");
                    exit();
                }
            }
            for(c=0;c<2;c++){
                wait( )
            }
        }
    }
    printf("MATHS");
}
```

1. How many times "CS" will be printed ?
2. How many times "DS" will be printed
3. How many times "Maths" will be printed ?
4. Create a process tree
5. How many total processes created (Including main () process)?